



SUBJECT	Lower Secondary Science Year 8
TOPIC	The Earth and its Atmosphere
ASSIGNMENT	Eight
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: _____

ASSIGNMENT GUIDANCE:

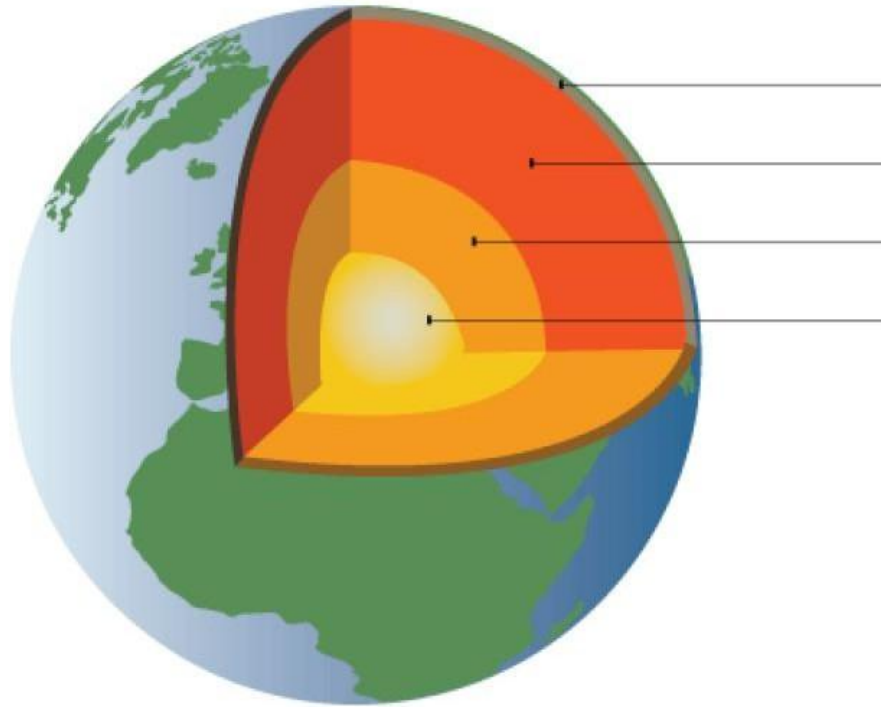
- You should complete this assignment by printing out this sheet and then handwriting in black ink, unless you have obtained specific permission to submit typed work. Then scan it in and submit it, as a single document.
- This is not a timed assignment. We estimate that this assignment may take you between 40 and 50 minutes for the assignment itself. However, if you need to take longer, then please do so.
- Make sure you write your full name clearly in the box at the top of this page.
- In order to complete this assignment, you will need a calculator, ruler, your coursebook and notes.



LS Science Assignment Eight

THE EARTH AND ITS ATMOSPHERE

1. Name the four layers of the Earth



(Total 4 marks)

2. Look at the descriptions of three rocks, A, B and C.

Rock A is formed in layers. It is soft and contains fossils.

Rock B is made when molten rock cools. It is hard and contains crystals.

Rock C is very hard. It contains distorted fossils due to high temperatures and high pressures.

- a. Complete the sentences to name each type of rock.

Choose from the list.

Rock A is _____ rock

Rock B is _____ rock

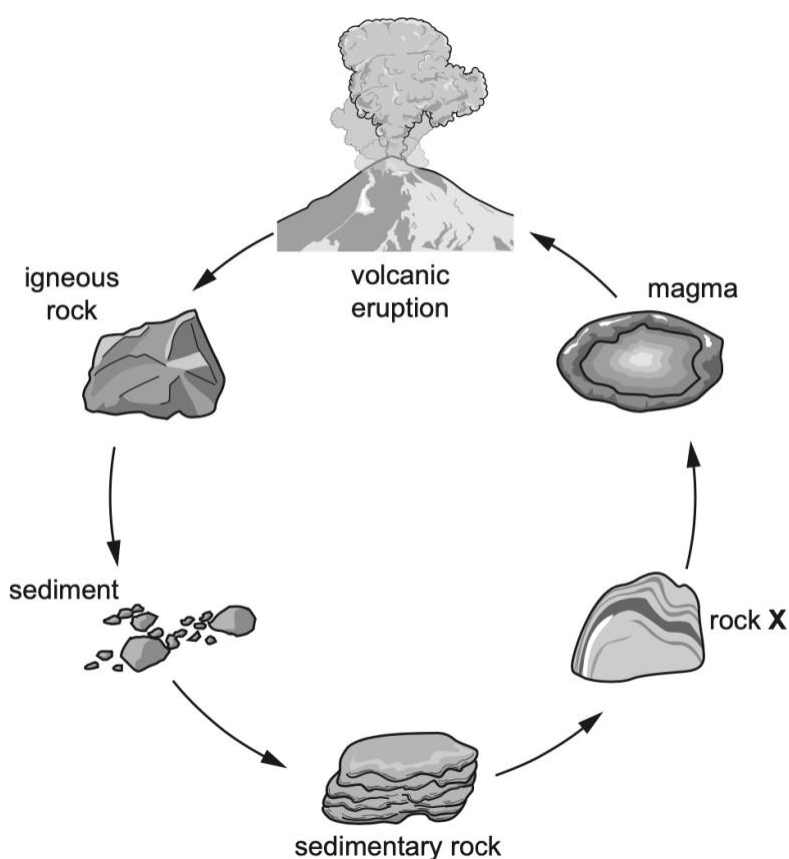
Rock C is _____ rock

(2 marks)

b. Describe how sedimentary rock is formed.

(2 marks)
(Total 4 marks)

3. The diagram shows different types of rocks and how they form.



a. Sedimentary rocks can be turned into rock X by heat and pressure.

What type of rock is X? _____

(1 mark)

- b. Which layer of the Earth contains magma?

Circle the correct answer.

atmosphere

inner core

mantle

outer core

(1 mark)

- c. Sedimentary rocks often contain the remains of dead animals and plants from millions of years ago.

What word is used to describe these remains?

(1 mark)

(Total 3 marks)

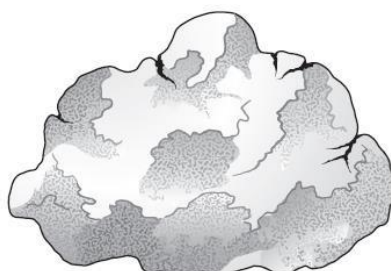
4. Helen weighed three pieces of rock and soaked them in water. The next day, she weighed them again. Her results are shown below.

rock	mass before soaking in water (g)	mass after soaking in water (g)
granite	26.3	26.3
marble	20.4	20.4
sandstone	25.5	27.6

- a. Rocks that have lots of small spaces are described as **porous**. What evidence is there in the table that sandstone is porous, but granite and marble are **not** porous?

(1 mark)

- b. Helen put the soaked sandstone into the freezer for 24 hours.



Water in the spaces in the sandstone froze and expanded.

- i. What would happen to the sandstone as the water froze and expanded?

(1 mark)

- ii. In the winter this process happens in rocks in the countryside. What is the name of this process?

(1 mark)

- c. Helen placed fresh pieces of granite, marble and sandstone in beakers of dilute sulphuric acid. The marble and sandstone reacted, while the granite did not.

Use Helen's results to explain why granite is more suitable than marble or sandstone for a statue in a **city centre**.

(1 mark)

- d.

- i. Draw a line from the name of each rock below to the group of rocks it belongs to.

(2 marks)

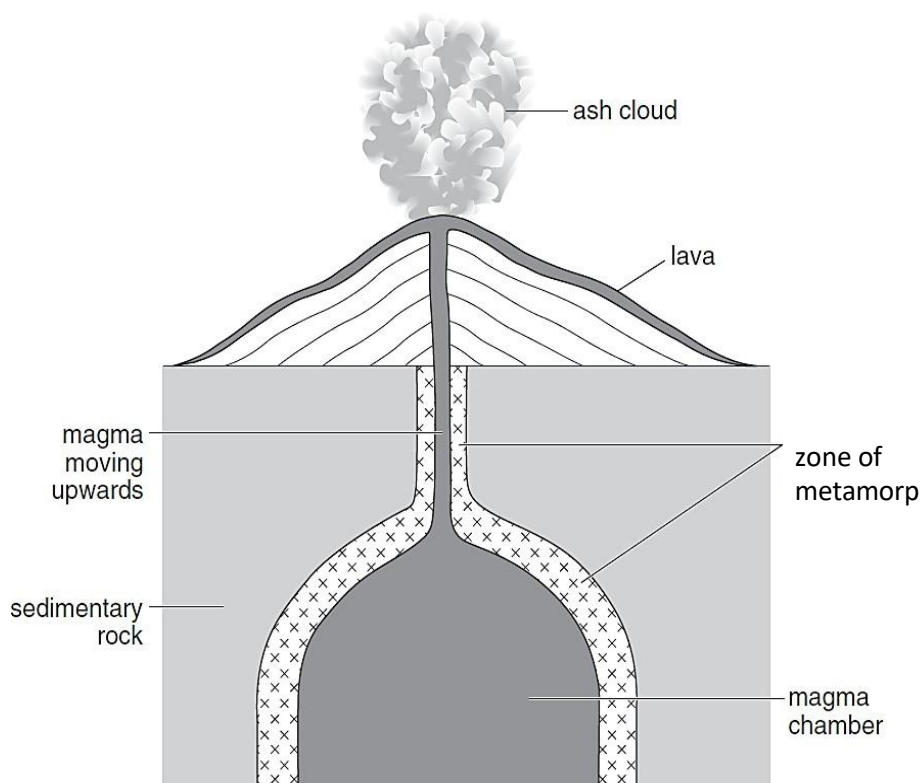
- ii. Draw a line from each group of rocks below to the way the group of rocks was formed.

(2 marks)

name of rock	group of rocks	way the group of rocks was formed
granite	sedimentary	the effect of high temperature and pressure on limestone
marble	igneous	formed when magma cools
sandstone	metamorphic	particles are deposited in layers

(Total 8 marks)

5. The diagram below shows a section through a volcano. Magma is moving up from a magma chamber. Some of the magma erupts to form lava. The liquid lava cools and becomes solid rock.



- a. Explain why magma deep underground stays liquid longer than lava on the surface.

(1 mark)

- b. As the magma cools underground, it solidifies, and crystals are formed.

- i. In what way will these crystals be different from the crystals formed when lava solidifies above ground?

(1 mark)

- ii. Give the reason for your answer.

(1 mark)

c. A zone of rock surrounding the magma has become a metamorphic rock.

i. What conditions would cause this to happen?

(1 mark)

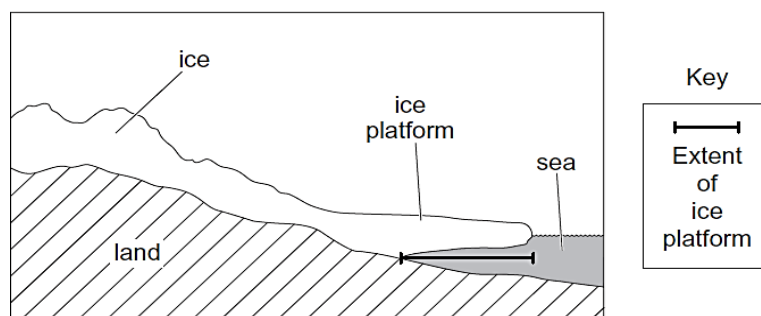
ii. Give the name of the metamorphic rock which is formed from limestone in this way.

(1 mark)

(Total 5 marks)

6.

a. Polar bears hunt seals from ice 'platforms'.



The table below shows the maximum distance that the ice platform extended beyond an island in the Arctic from 1950–2010.

Year	Extent of ice platform/km
1950	3.5
1960	3.25
1970	2.8
1980	2.6
1990	2.4
2000	2.3
2010	2.2

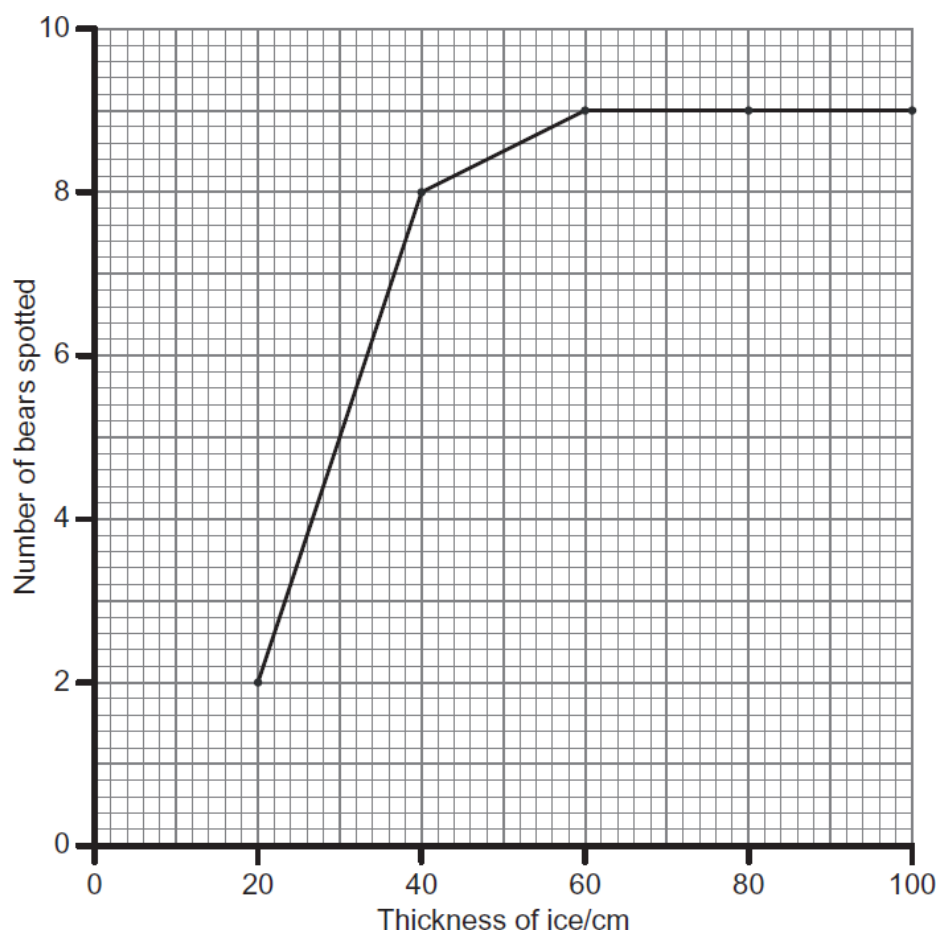
i. State the trend shown by the information given in the table.

(1 mark)

- ii. It is thought that the change in the ice platform is due to global warming. Explain fully the cause of global warming and how this information shows evidence that it is happening.

(3 marks)

- b. Scientists also found that the platforms are getting thinner and this has affected the number of polar bears on the ice. The graph below shows the number of polar bears spotted on different thicknesses of ice.



- i. Suggest a reason why no polar bears have been spotted on ice below 20cm in thickness.

(1 mark)

- ii. Suggest what will happen to polar bears if the ice thickness continues to decrease.

(1 mark)
(Total 6 marks)

7. The table below shows some waste materials produced in Belfast in one year.

Material	Waste produced/ tonnes
paper	16 500
plastic bottles	4 500
glass	6 000
food and garden waste	33 000
aluminium cans	3 500
cardboard	7 500

- a. In Belfast 50% of the paper waste is recycled. Calculate the number of tonnes of paper waste that is recycled in one year (show your working out).

_____tonnes
(2 marks)

- b. Give **one** advantage of recycling.

(1 mark)
(Total 3 marks)

8. This question is about tectonic plates.

a. Tectonic plates float on molten magma in the mantle.

i. Write down the name of the layer of the Earth made of tectonic plates.

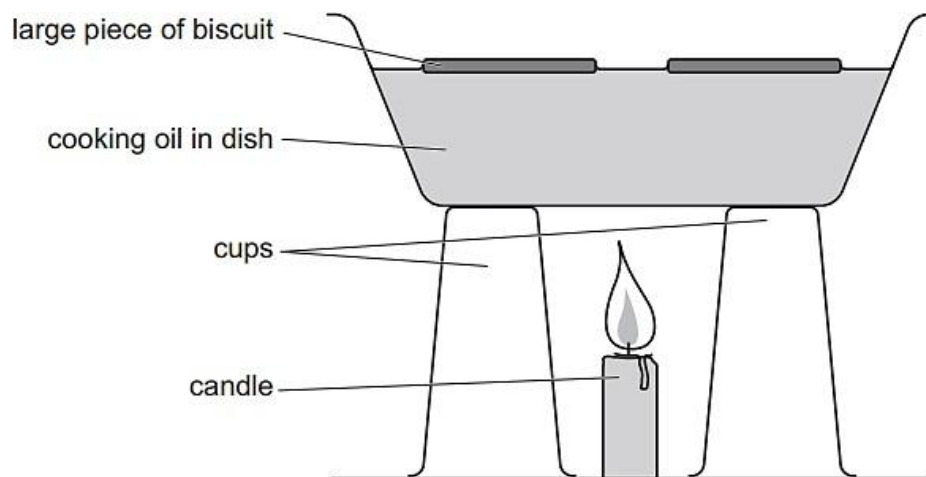
(1 mark)

ii. Explain why tectonic plates float on the mantle.

(1 mark)

b. Yuri models what happens at the boundary between two tectonic plates.

Look at the diagram of Yuri's model.



Each large piece of biscuit models a tectonic plate.

The large pieces of biscuit float on the cooking oil.

The cooking oil models the magma in the mantle.

i. Which part of the Earth does the lighted candle in the model represent?

(1 mark)

ii. Predict what will happen to the pieces of biscuit.

Explain your answer.

prediction

Explanation

(2 marks)

(Total 5 marks)

TOTAL FOR ASSIGNMENT 38 MARKS